Techforce Services

Kimai Cloud Migration Project

GitHub Repository:

This project is maintained on GitHub:

Repo Name: Cloud-Migration-Project

Link: https://github.com/angelinedev/Cloud-Migration-Project

Project Structure:

- **terraform/** – Infrastructure-as-Code setup using Terraform, with a modular design and remote backend configured in AWS S3.
- **kimai/** – Docker configuration for Kimai, including a Dockerfile and docker-compose.yml for containerized deployment.
- **Jenkinsfile** – Pipeline as Code for Jenkins, automating build, test, and deployment processes.
- **docs/** – Documentation folder containing High-Level Design (HLD) and Low-Level Design (LLD) for the project architecture and implementation details.

Tech Stack

Component	Tech
Backend	PHP Symfony Framework)
Frontend	HTML/CSS/JS Bundled)
DB	MariaDB
Server	Nginx
Runtime	PHP FPM
OS	Amazon Linux 2 EC2

IaC	Terraform
Containerization	Docker
CI/CD	Jenkins
Monitoring	CloudWatch, Grafana
Logging	CloudWatch Logs
Access Control	AWS IAM, Bastion Host

Deployment Architecture:

Textual Architecture Diagram

```
[ User ]
   |
[ Load Balancer (Public) ] [ WAF ]
   |
[ Bastion Host (Public Subnet) ]
   |
[ VPC ]
   ├── EC2 (Kimai + Docker) [Private Subnet]
   └── RDS (MySQL DB)       (Private Subnet]
```

Security Highlights:

- Bastion host for secure SSH
- IAM with least privilege
- Security Groups with only required ports open
- ALB protected by AWS WAF

Repository Structure:

```
Cloud-Migration-Project/
 ├─ terraform/
|    ├─ main.tf
|    ├─ variables.tf
|    ├─ outputs.tf
|    └─ modules/
|     └─ ec2/
|         ├─ main.tf
|         ├─ variables.tf
|         └─ outputs.tf
 ├─ Jenkinsfile
 ├─ kimai/
|      ├─ Dockerfile
|      └─ docker-compose.yml
 ├─ docs/
|    ├─ HLD.md
|    └─ LLD.md
 └─ README.md
```

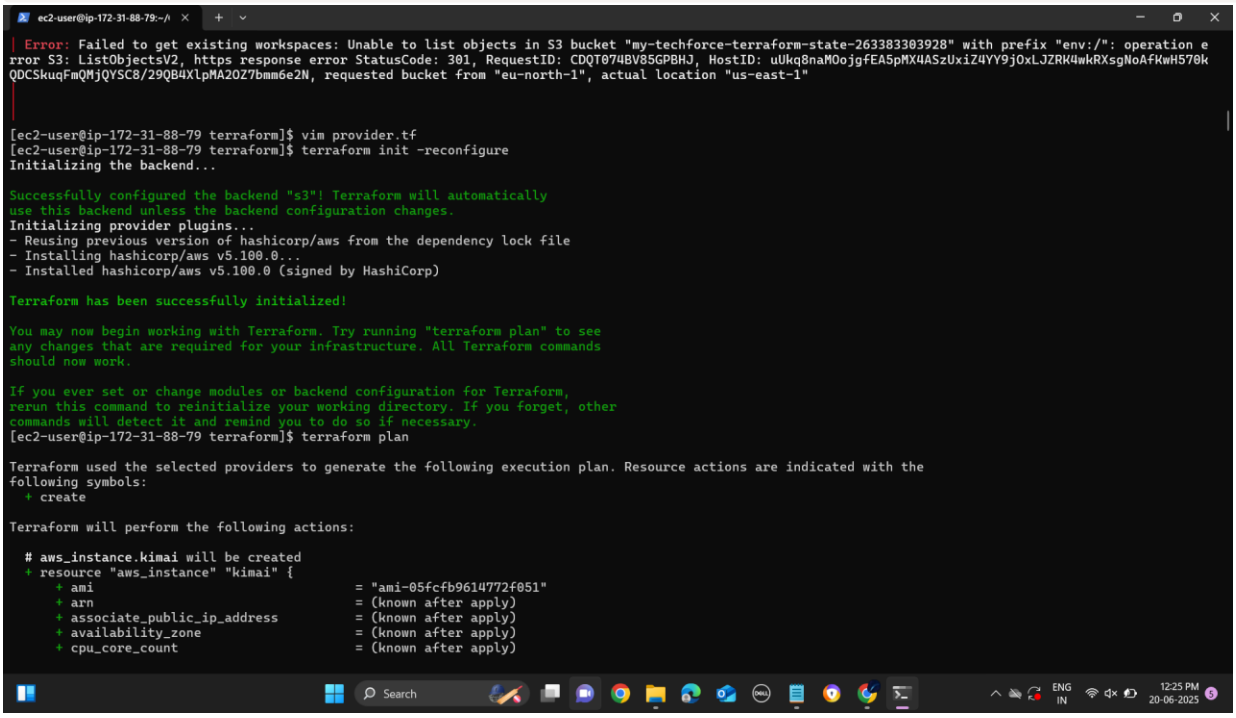
Terraform Setup:

All Terraform modules and configuration files are located in the terraform/ directory inside the GitHub repository.

- Modular setup for reusability
- Remote state managed in S3
- Uses t3.large for Kimai EC2 and t2.micro for Bastion Host
- Outputs public/private IPs, instance IDs To

deploy:

```
cd terraform
terraform init
terraform apply -auto-approve
```



Docker Setup:

Kimai is containerized using a multi-stage Dockerfile to ensure efficient and lightweight builds.

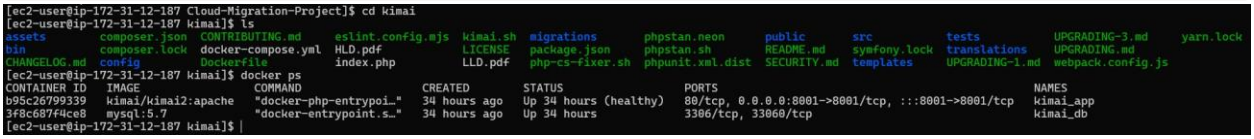
The container is:

- **Non-root**, enhancing security.
- Equipped with a **healthcheck** for better reliability.
- Fully **production-ready**.

To build the image locally,

```
docker build -t kimai-app .

docker run -p 80:8001 kimai-app
```



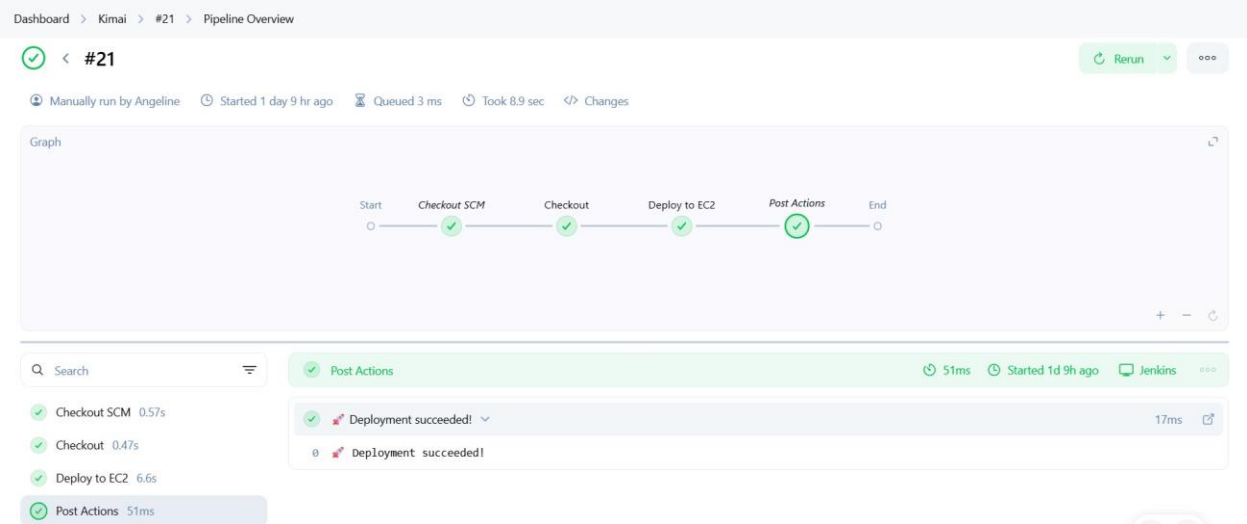
CI/CD Pipeline (Jenkins):

Pipeline is stored as Jenkinsfile in the GitHub repository.

Trigger: Push to main branch

- Checkout
- Build Docker Image Run
- Tests
- Push to DockerHub
- Deploy via SSH to EC2

Pipeline is fully automated using a Jenkinsfile .



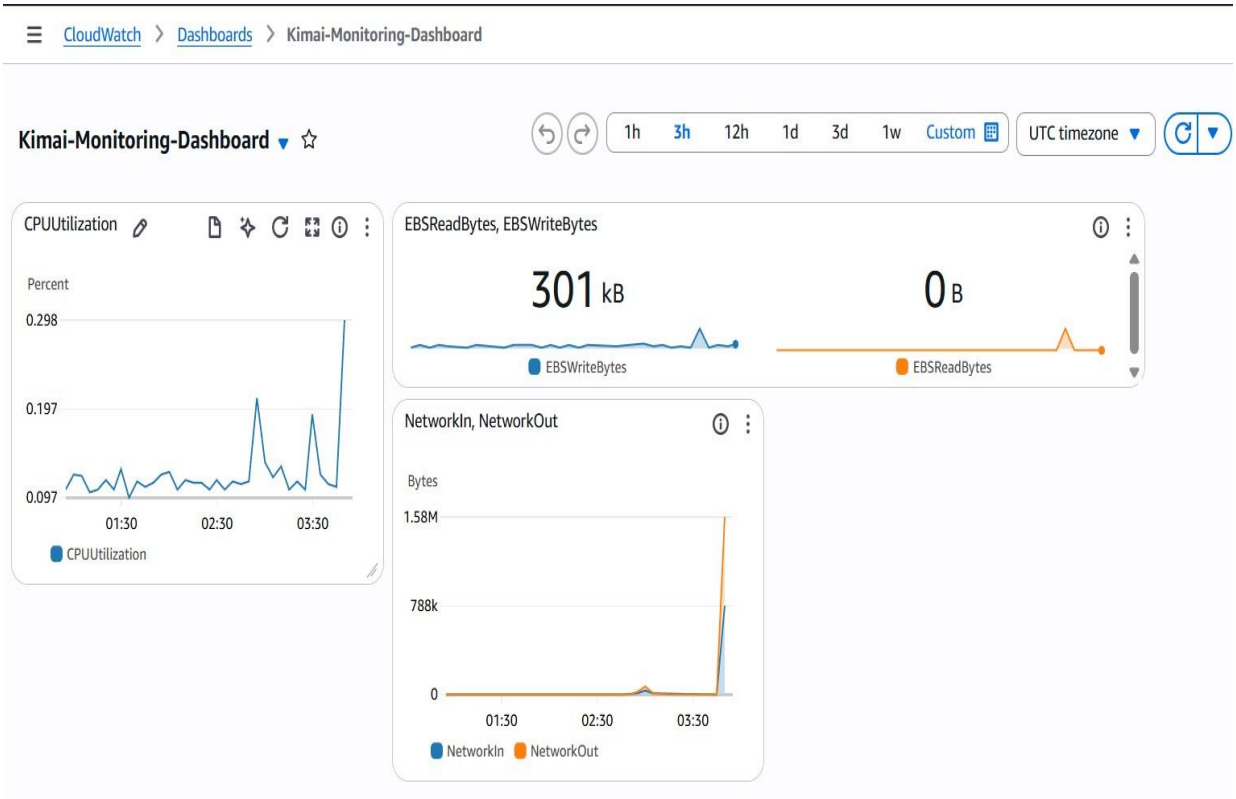
Security:

- WAF filters traffic on Load Balancer
- Bastion Host enables safe key-based access to private instances
- IAM Role attached to Kimai EC2 for CloudWatch agent and S3 access



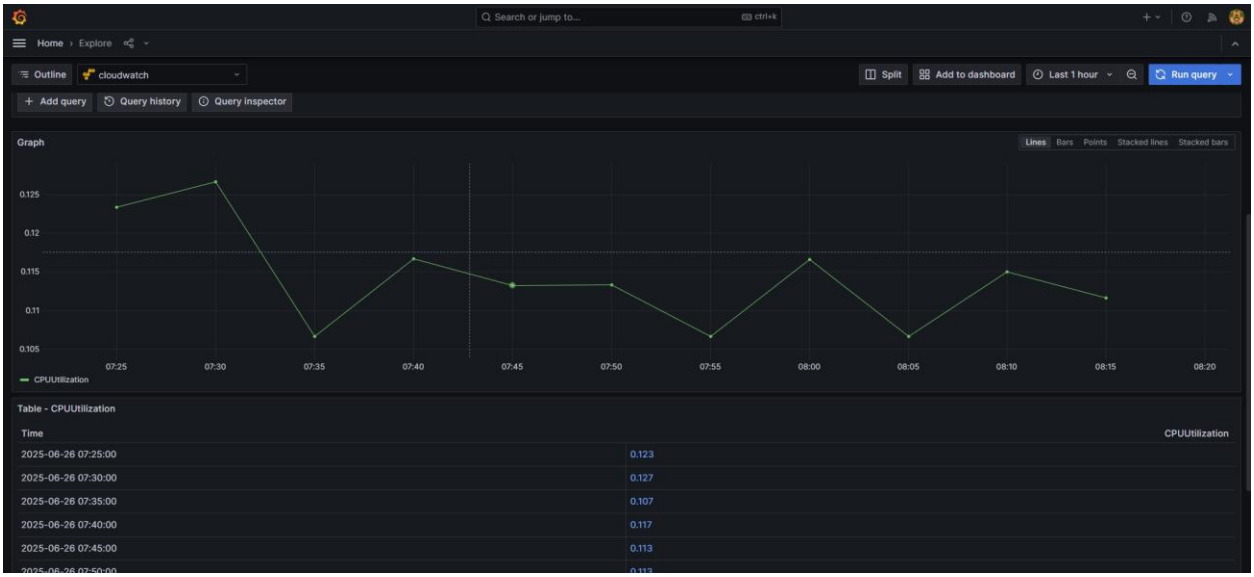
Monitoring&Logging

- CloudWatch Agent installed and configured on EC2 instance
- Monitors:
  - CPU usage
  - Memory and Disk space
  - Application logs
- Alerts configured for:
  - High CPU ⍰⍰ 80%⍰
  - Health Check failures



Grafana Setup:

- Grafana is installed on an EC2 instance.
- Integrated with AWS CloudWatch using an IAM role for secure access
- Dashboards are configured to
  - CPU Utilization
  - Memory Usage
  - Disk Usage
  - Network Traffic
- Alerts are configured with defined



Cost Estimation (Monthly):

Resource	Cost
EC2 (t3.large)	$33.28
Bastion Host (t2.micro)	$7.62
EBS 28GB total)	$2.80
S3 (backups + state)	$0.12
CloudWatch	$2.00
Total	**$45 48/month**

Final Checklist:

- ☑ High-Level & Low-Level Design in /docs
- ☑ Terraform infra with S3 remote state
- ☑ CI/CD with Jenkins
- ☑ Dockerized Kimai app
- ☑ Secure via Bastion IAM WAF
- ☑ Monitoring and alerts with CloudWatch Grafana
- ☑ Cost Estimate and Final Report generated
- ☑ All code and documentation are available in the GitHub repository:

https://github.com/angelinedev/Cloud-Migration-Project